OPTIMIZING DISCOUNTS FOR MAXIMUM PROFITABILITY

Agenda

- Problem Statement
- Research Questions
- Data Overview
- Methodology
- Descriptive Analysis Insights
- Correlation Analysis Insights
- Discount Efficiency Analysis
- Recommendations
- Final Conclusion

Problem Statement

- **Analyze Discount Impact :**
  Examine how different discount levels affect profitability across regions, product categories, and transaction volumes.

- **Identify Profit Erosion Thresholds :**
  Determine the discount levels at which profits start declining or turning negative.

- **Recommend Optimal Discount Strategies :**
  Suggest discount strategies that maximize profit while maintaining sales performance.

Research Questions

- **Are there cases where high discounts do not significantly increase sales?**
- **Which regions or categories show inefficient discounting?**

Data Overview

This dataset contains transactional retail data capturing customer purchases across different regions, product categories, and pricing conditions. It includes key metrics such as sales, profit, discount levels, and quantity, enabling analysis of how pricing strategies impact business performance.

Relevant Columns Use for This Research :

- **Region : To compare performance across geographical areas**
- **Category / Sub-Category : To analyze product-level behavior**
- **Discount : Key variable (pricing strategy)**
- **Quantity : Demand indicator**
- **Sales : Revenue generated**
- **Profit : Final business outcome**

	Region	Category	Sub-Category	Discount	Unit Price	Quantity	Sales	Profit
0	South	Books	Non-Fiction	5	36294	2	68958.6	10525.09
1	North	Groceries	Rice	20	42165	1	33732.0	6299.66
2	East	Kitchen	Juicer	20	64876	4	207603.2	19850.27
3	East	Groceries	Oil	15	37320	5	158610.0	36311.02
4	West	Clothing	Kids Wear	10	50037	1	45033.3	9050.04

Methodology

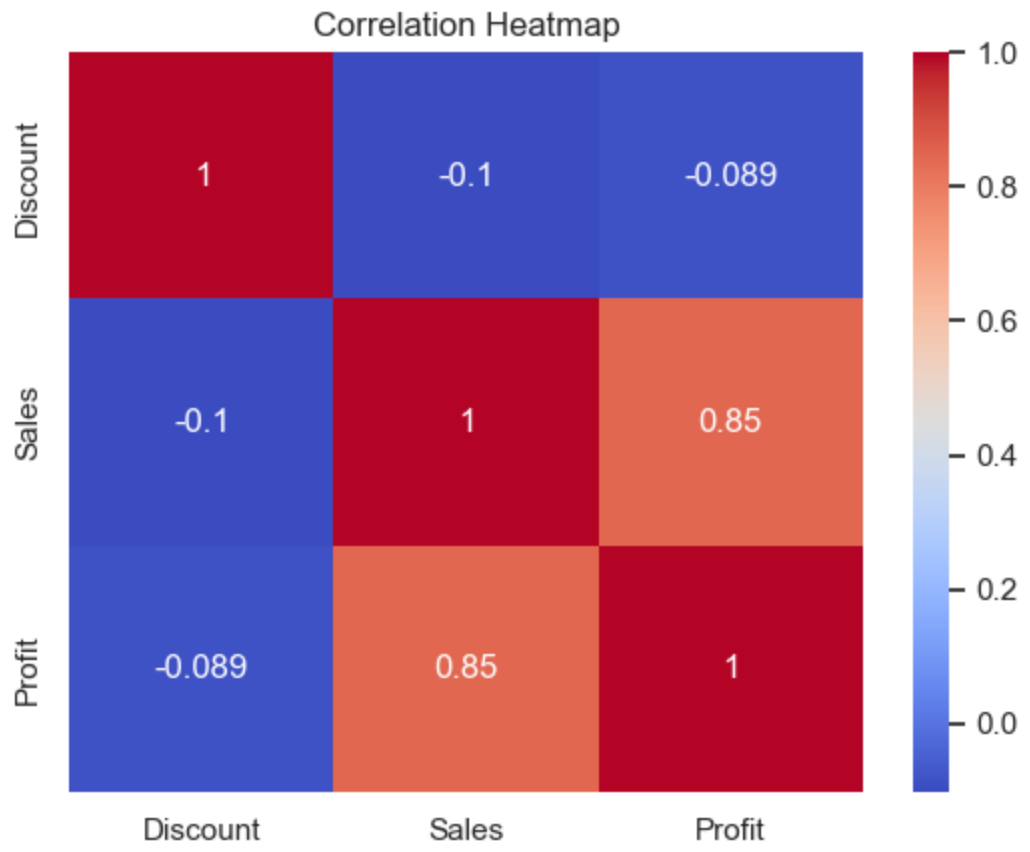
Steps	Methodology Stage	Key Activities
1	Data Preparation	Cleaned data, handled missing values, selected relevant columns
2	Data Understanding	Analyzed summary statistics for Sales, Profit, Discount, Quantity
3	Relationship Analysis	Studied impact of Discount on Sales and Profit using correlation & plots
4	Discount Evaluation	Compared performance across discount levels to identify inefficiencies
5	Insights & Recommendations	Derived business insights and suggested optimal strategies

Descriptive Analysis Insights

- **The average discount offered is approximately 10%, indicating a moderate pricing strategy.**
- **Discounts range from 0% to 20%, with no extreme discounting observed.**
- **All transactions are profitable, with no negative profit values.**
- **There is high variability in sales and profit, suggesting inconsistent performance across transactions.**
- **The dataset shows potential to analyze how discounts influence profitability variation.**

Correlation Analysis Insights

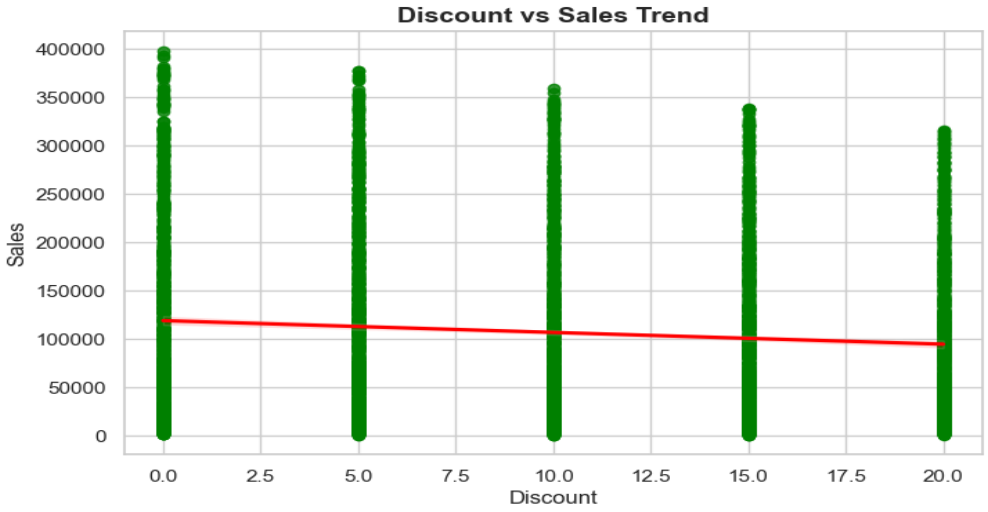
- **The correlation between discount and sales (-0.10) is very weak, indicating that higher discounts do not significantly increase sales.**
- **The relationship between discount and profit (-0.08) is also weak and negative, suggesting that discounts slightly reduce profitability.**
- **A strong positive correlation (0.85) exists between sales and profit, showing that increasing sales directly improves profitability.**
- **These findings indicate the presence of inefficient discounting, where discounts are not effectively driving revenue growth.**



Discount Efficiency Analysis

Impact of Discounts on Sales & Profit

- Sales and profit both decline as discount levels increase, indicating that higher discounts do not improve performance.
- The highest sales and profit are observed at 0%–5% discount levels, suggesting that heavy discounting is unnecessary.
- Discounts do not significantly influence customer purchase behavior.
- Across categories, discount levels remain similar (~10%), but profitability varies, highlighting an inefficient uniform discount strategy.



Inefficient Discounting Across Regions & Categories

Regional Insights:

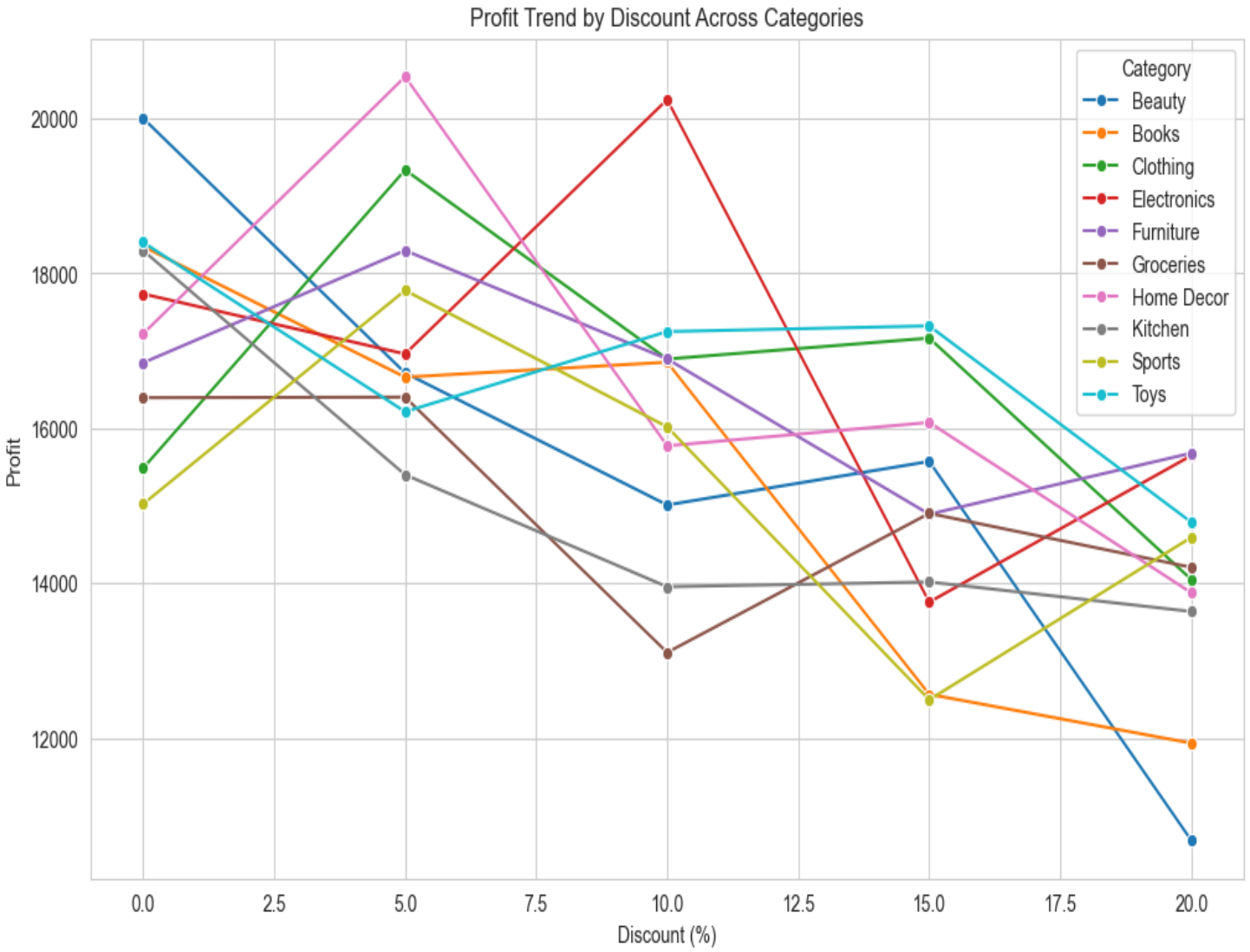
- All regions show declining profit with increasing discounts
- Profit drops significantly beyond 10% discount
- No corresponding increase in sales or quantity observed

Category Insights:

- Several categories show strong profit erosion at higher discounts (Beauty, Books, Kitchen, Sports)
- Limited evidence that higher discounts improve performance

Critical Finding:

- Discounts above 15% are consistently inefficient
- Leads to profit loss without sales gain



Identifying Optimal Discount Threshold

Key Findings:

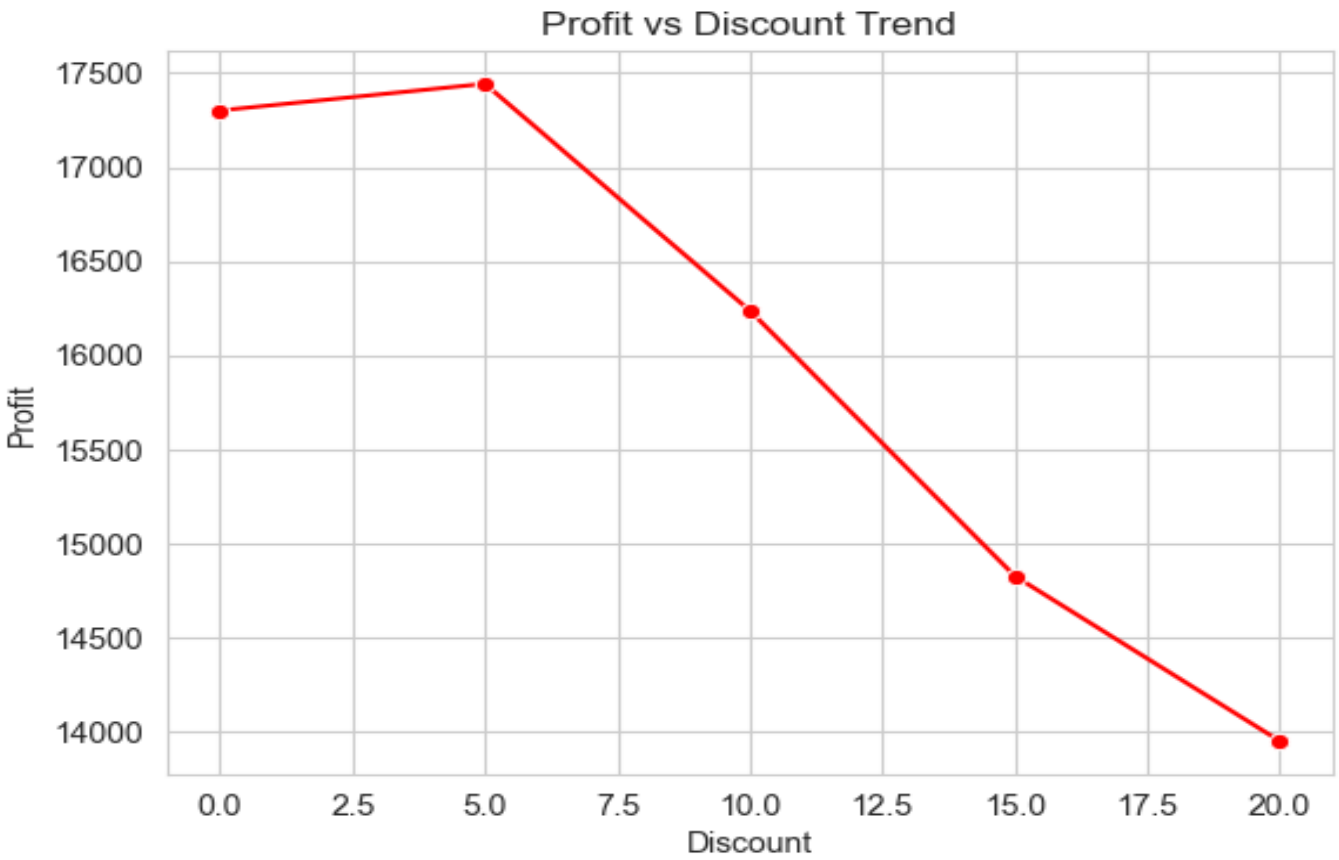
- **Profit peaks at 5% discount (~17.4K)**
- **Profit declines steadily beyond 5%**
- **Significant drop Observed after 10% discount**

Trend Insights:

- **10% → noticeable decline in profitability**
- **15%–20% → sharp profit erosion**
- **Sales also decrease with higher discounts**

Critical Threshold:

- **Optimal discount level: 5%**
- **Profit erosion begins beyond: 10%**



Recommendations

Standardize Discount Policy:

- Limit discounts to ≤ 10%
- Avoid high discounting (15%–20%) across all regions

Adopt Optimal Discount Level:

- Implement ~5% discount as default strategy
- This level delivers maximum profitability

Eliminate Inefficient Discounting:

- Stop offering discounts where:
- Sales do not increase
- Profit declines

Region-Based Control:

- Monitor regions applying higher-than-average discounts
- Enforce controlled discounting policies

Category-Level Optimization Reduce discounts in:

- Beauty, Books, Kitchen, Sports

Expected Business Impact

- Increase overall profitability
- Reduce revenue leakage
- Improve pricing efficiency
- Enable data-driven discount decisions

Final Conclusion

Current discount strategies are ineffective — reducing discounts to optimal levels (around 5%) can significantly improve profitability without impacting sales performance.

THANK YOU